Tagalog Retention and English Acquisition under Equal-Budget nanochat Continual Pretraining v1.1

A Preregistered Post-P2 Reverse-Direction Study on WikiText-TL-39 and WikiText-103

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Abstract

When a fresh Tagalog-pretrained nanochat parent, whose Tagalog provenance has passed a filed parent-eligibility gate, is continued for a matched token budget on English WikiText-103 raw, does held-out Tagalog bits-per-byte (BPB) rise more than under matched extra-Tagalog continuation, and does held-out English BPB fall? This question was preregistered as AsPredicted #307342 before any P3 tokenizer, parent, child branch, validation BPB, or test BPB existed. P3 was designed after P2 unblinding; it is therefore a prospectively preregistered post-P2 reverse-direction study, not an outcome-independent confirmation of P2 and not an amendment to AsPredicted #306780 or #306935. The study trains fresh P3 Tagalog TL0 parents from WikiText-TL-39 train documents using a new P3 train-only 32,768-merge BPE tokenizer, sequence length $T = 2048$, and Karpathy's nanochat base trainer pinned at commit `92d63d4`. After the locked d8+d20 P0-T eligibility gate, the final d20 parent B0 is frozen. Matched d20 continuations compare B1 extra Tagalog, B2 English WikiText-103 raw, and a pre-frozen B3 50/50-document tradeoff stream for $D_{\text{phase2}} = 19{,}267{,}584$ tokens ($N = 294$, $B = 65{,}536$). Primary outcomes are locked full-split Tagalog and English `val_bpb_full` after final child checkpoints. The formal Gate X release reports $C_{\text{tl}} = \text{TL}(B2) - \text{TL}(B1) = 1.023484$ (filed ≥ 0.01: **observed**) and $G_{\text{en}} = \text{EN}(B2) - \text{EN}(B1) = -1.697955$ (filed ≤ -0.01: **observed**). One authorized B2-only secondary legacy-holdout event yielded English 1.357842 and Tagalog 2.493197 BPB; B1 and B3 were not tested. The contribution is a locked equal-exposure reverse-direction continual-pretraining measurement in one seed, not a claim of universal catastrophic forgetting, a chat/SFT result, a CORE benchmark result, or evidence that B3 mitigates forgetting.

Keywords: Tagalog retention; English acquisition; continual pretraining; WikiText-TL-39; WikiText-103; bits-per-byte; equal exposure; nanochat; preregistration; reverse-direction; post-P2.

arXiv categories: cs.CL, cs.LG.

Registration and deposits: AsPredicted #307342 (https://aspredicted.org/wd2pc8.pdf); ResearchBox #8834 (https://researchbox.org/8834); AsCollected https://ascollected.org/F36_C2C; code https://github.com/pageman/nanochat-filipino; Hub https://huggingface.co/pageman/nanochat-filipino-p3-tl-then-en. Run ID: `p3-20260819T192700Z-92d63d4`.

1 Introduction

Pajo [1] measured equal-exposure Tagalog BPB across nanochat depths on WikiText-TL-39 (AsPredicted #306780). Pajo [2] then asked the forward continual-pretraining question: after an English WikiText-103 parent clears an English eligibility gate, does matched Tagalog continuation raise English held-out BPB more than extra English, and does Tagalog BPB fall (AsPredicted #306935)? That forward study is complete. It does not answer the reverse arrow.

P3 asks the reverse causal question under the same instrument family: after a *fresh* Tagalog parent clears a Tagalog eligibility gate, does matched *English* continuation raise Tagalog held-out BPB more than extra Tagalog, and does English held-out BPB fall? The study was designed after P2 Gate U/V unblinding on 19 August 2026. It is therefore a prospectively preregistered **post-P2** reverse-direction study [5]. It is not an outcome-independent confirmation of P2 and does not amend #306780 or #306935.

The confirmatory question, locked before any P3 tokenizer, TL0, B0–B3, validation BPB, or test BPB existed, is:

> After a newly trained P3-specific Tagalog nanochat parent (TL0) passes P0-T, does English continuation raise Tagalog held-out BPB more than matched extra Tagalog, and does English held-out BPB fall?

Filed directional predictions at depth 20:

$$C_{\mathrm{tl}} = \mathrm{TL}(B2) - \mathrm{TL}(B1) \geq 0.01, \tag{1}$$

$$G_{\mathrm{en}} = \mathrm{EN}(B2) - \mathrm{EN}(B1) \leq -0.01. \tag{2}$$

B3 is a predeclared 50/50-*document* trade-off arm, not mitigation.

This paper contributes: (i) a locked reverse-direction equal-budget nanochat run with fresh P3 weights and tokenizer; (ii) P0-T before any English child token; (iii) six child validation cells plus descriptive B0 English, sealed before one B2-only secondary test touch; (iv) a Gate X release that reports both primary filed patterns as **observed** in this one-seed apparatus.

2 Related work

2.1 Filipino resources and prior nanochat filings

Cruz and Cheng constructed WikiText-TL-39 [6]. P1.1 used a reconstructed-article 70/15/15 split and selected $D^* = 20$ by exact minimum `val_bpb_full` [1, 3]. P2 reversed the usual "Tagalog base then English" folklore by starting from English [2, 4]. P3 closes the remaining arrow without loading P1.1 or P2 weights as parents.

2.2 Continual pretraining and forgetting

Sequential interference is classical [12, 13]. LLM work often continues an English or multilingual base on a new language or task [10, 11, 9]. P3 stays inside WikiText-family text, nanochat depth dial, and BPB rather than CORE or chat [8]. Equal token budgets follow P1.1/P2 so stream identity is not confounded with compute [14, 15, 16].

2.3 Evaluation

WikiText established clean Wikipedia LM evaluation in English [7]. BPB remains comparable across vocabularies [17, 18]. Official P3 BPB is mean token NLL divided by $\ln 2$ times mean UTF-8 bytes per token, full split, $T = 2048$, BOS-best-fit. In-loop trainer BPB is not confirmatory.

2.4 Preregistration

Preregistration reduces undisclosed flexibility [19, 20]. AsPredicted #307342 locked the reverse question, P0-T, arms, budgets, cutoffs, one B2-only test touch, and post-P2 disclosure before P3 outcomes existed [5].

3 Glossary

P1.1 / P2
Prior filings #306780 / #306935. Not amended. Not P3 parents.

TL0
Fresh Tagalog-from-scratch P3 parent (d8 and d20).

P0-T
Tagalog eligibility: TL0 beats untrained and Tagalog-train add-1 byte-unigram floors by ≥ 0.01 BPB at both depths before English child tokens.

B0
Frozen TL0 d20 parent; no additional train tokens at freeze.

B1
Extra-Tagalog continuation, D_{phase2} (active control).

B2
English WikiText-103-raw continuation, D_{phase2} (intervention). Only tested arm.

B3
Pre-frozen 50/50-*document* EN/TL mix, D_{phase2}. Trade-off, not mitigation.

C_{tl}
Tagalog `val_bpb_full`(B2) − `val_bpb_full`(B1). Filed ≥ 0.01.

G_{en}
English `val_bpb_full`(B2) − `val_bpb_full`(B1). Filed ≤ -0.01.

val_bpb_full
Official full-split validation BPB after the final checkpoint. Primary DV.

test_bpb
Same formula on legacy external holdouts. Secondary. B2 only, after val seal.

D_{phase2}
$294 \times 65{,}536 = 19{,}267{,}584$ model-visible tokens.

T Context length, frozen at 2048.

4 Methods

4.1 Registration and authority

AsPredicted #307342 governs.

- PDF SHA-256:
  `6cfad0386dff689ad73fa2bf80b70dd4ad191dc44e21e3e4c11c06825ae550b1`
- Protocol SHA-256 at filing:
  `899ba83f0b36f2b4bf4c16b3c675e58788d7763cb439f8a8c3a3c061bda2b986`

ResearchBox #8834 is the deposit. AsCollected records public-data provenance at `https://ascollected.org/F36_C2C`. Authority: filed PDF $\gg$ protocol $\gg$ `LOCK.json` $\gg$ dated deviations $\gg$ this manuscript. The study does not amend #306780 or #306935.

4.2 Pin and pipeline

nanochat commit `92d63d4e8bb4df75c3b71618f31ddde2378b2bcd`. Pipeline: `tok_train` $\rightarrow$ `base_train` / `continue_from_frozen` $\rightarrow$ `scripts/p3/evaluate_bpb.py`. CORE off. No Hugging Face `Trainer`. No attention patch relative to the pin. No P1.1/P2 weights as parents.

4.3 Corpora and splits

Tagalog train/val/test documents are the frozen P1.1 reconstructed-article 70/15/15 manifests (train SHA-256 prefix `2b0474c5`; val `4d51644b`; test `3bd19345`). English uses official WikiText-103-raw Merity splits from Hugging Face `Salesforce/wikitext`, config `wikitext-103-raw-v1`, revision prefix `b08601e0` (val document manifest `874dec29`; test `2bccabc0`). Cleaning: LF normalization; drop null/empty or documents $>$ 200,000 characters; stop if drops $>$ 5%. Tests stay outside train/tokenizer mounts.

4.4 Tokenizer

A new P3 Tagalog 32,768-merge BPE is trained on Tagalog *train* only (tokenizer SHA-256 prefix `04436b85`; full hash in Appendix A). The same tokenizer is used for all arms and both languages' confirmatory BPB.

4.5 TL0, P0-T, and B0

TL0 is trained at depths 8 and 20 from fresh weights on Tagalog train with $T = 2048$, $B = 65{,}536$, and ceiling budget $N_{\mathrm{TL0}} = \lceil 3T_{\mathrm{tl\_train}}/65536 \rceil = 294$. P0-T requires both depths to beat untrained and add-1 byte-unigram floors by ≥ 0.01 BPB on Tagalog val before any English child token. On PASS, freeze final d20 as B0 (checkpoint SHA-256 prefix `ae621be2`).

4.6 Mechanical carry-forward of budgets and cutoffs

The phase-two step count $N = 294$, contrast cutoff 0.01 BPB, and B3 50/50-*document* mix recipe were carried forward as filed post-P2 *mechanics* matching the P2 apparatus (same B, T, and equal-budget design language). They were not tuned to P2 contrast magnitudes, signs, or Gate V test numbers. P3 remains a prospective filing of the reverse arrow after P2 unblinding, not a magnitude-matched replication of P2.

4.7 Phase-two arms

From identical B0 weights, with `load_optimizer=False` and fresh Muon+AdamW, peak LR $= 0.3\times$ TL0 peak, warmup 14:

- B1: Tagalog stream only (`p3-b1-extra-tl-d20`, SHA prefix `3f98784b`).
- B2: English stream only (`p3-b2-en-d20`, SHA prefix `5ee34b20`).
- B3: pre-frozen 50/50-document mix (seed 42, $K = 28472$, frozen before TL0 validation; checkpoint SHA prefix `521bea16`; mix-order SHA prefix `b6ae432b`).

Each arm runs exactly $N = 294$ new steps ($D_{\text{phase2}} = 19{,}267{,}584$). Terminal checkpoint only; no mid-run selection.

4.8 B3 document mix versus realized exposure

B3 is defined on *documents*, not tokens. Gate E froze a 50/50 EN/TL document mix ($K = 28472$ per language after SHA-sorted selection, seed 42) before TL0 validation. Because English Wikipedia documents are longer on average, realized UTF-8 byte shares in the frozen mix are heavily English-skewed (Table 1). BPE token shares were not a Gate E seal field; byte shares are the sealed exposure disclosure.

Table 1: B3 exposure: documented mix rule versus Gate E realized UTF-8 byte shares.

Quantity	English	Tagalog
Document share (filed mix rule)	0.50	0.50
UTF-8 byte share (realized; Gate E)	≈ 0.961	≈ 0.039
UTF-8 bytes in mix (Gate E)	539,903,397	21,726,972

Mix-order SHA-256 prefix `b6ae432b`; full hash in Appendix A. B3 remains a trade-off coordinate, not a mitigation claim.

4.9 Gate S recovery

Early Gate S attempts failed when English training data were missing or corrupted (including AppleDouble junk), yielding a non-official zero-step/partial path. The official disposition, recorded in the Gate X preflight, was: quarantine the partial attempt; clean restart from frozen B0 with fresh optimizer on a cleaned English staging path (`en-clean`); complete exactly 294 steps with exit code 0. No result-informed change of arms, budgets, cutoffs, or metrics—no result shopping. The quarantined partial is not an official B2 outcome.

4.10 Evaluation and access control

Gate U seals six child cells B1/B2/B3 $\times$ {TL, EN} plus one descriptive B0 English `val_bpb_full`, with `test_access`=0 at seal. B0 English is excluded from contrasts. After the seal, Gate V performs one authorized B2-only event with two legacy external holdouts (English WT103-raw test manifest; Tagalog P1.1 `test.jsonl`), then `test_access`=1. B1 and B3 are not tested. Gate X is a status-only preflight plus one-time release of the sealed package. Raw test text remains restricted.

4.11 Compute note

TL0 d8 and d20 both completed $N = 294$ steps on a Runpod A40 (`bef5h2lzy6f3mp`). Wall-clock hours were not sealed as confirmatory fields in the Gate I receipts and are omitted here.

5 Results

5.1 Primary sealed validation table

Table 2 reports full-split `val_bpb_full` after final checkpoints (one seed; point estimates only).

Table 2: Primary P3 validation cells (Gate U seal). Lower BPB is better. B0 English is descriptive only.

Arm	Role	Tagalog `val_bpb_full`	English `val_bpb_full`
B0	Frozen TL0 d20 parent	(via P0-T; not recomputed at U)	2.618891
B1	Extra Tagalog	1.468600	3.032277
B2	English continuation	2.492084	1.334322
B3	50/50-document mix	1.193565	1.348593

5.2 Registered contrasts

$$C_{\mathrm{tl}} = 2.492084 - 1.468600 = 1.023484 \geq 0.01 \qquad \textbf{observed}, \tag{3}$$

$$G_{\mathrm{en}} = 1.334322 - 3.032277 = -1.697955 \leq -0.01 \qquad \textbf{observed}. \tag{4}$$

Trade-off contrasts (not success criteria; B3 is not mitigation):

$$C_{\mathrm{tl}}(B3) = \mathrm{TL}(B3) - \mathrm{TL}(B1) = -0.275035, \tag{5}$$

$$G_{\mathrm{en}}(B3) = \mathrm{EN}(B3) - \mathrm{EN}(B1) = -1.683684. \tag{6}$$

In this apparatus, English continuation raised Tagalog BPB relative to extra Tagalog by more than the 0.01 cutoff, and lowered English BPB relative to extra Tagalog by more than the 0.01 cutoff. B3 improved Tagalog relative to B1 while remaining near B2 on English; that pattern is reported as a trade-off coordinate, not as mitigation of the primary contrast.

5.3 Secondary B2-only tests

After the validation seal, one authorized Gate V event on B2 only scored English test BPB 1.357842 and Tagalog legacy-holdout BPB 2.493197 under the P3 Tagalog BPE. These are secondary legacy/external holdout outcomes. They are not a test-set $C_{\mathrm{tl}}/G_{\mathrm{en}}$, do not alter the sealed contrasts, and must not be confused with P1.1's native-BPE `test_bpb` or with P2 Gate V numbers (explicit non-reuse in Section 6.3).

6 Discussion

6.1 What is claimed

Within this one-seed, fixed-parent, fixed-budget, P3-tokenizer apparatus, both filed primary directional patterns were observed. The study supports a narrow causal claim about the registered English-versus-extra-Tagalog continuation streams after a P0-T-cleared Tagalog parent.

6.2 What is not claimed

The paper does not claim a universal catastrophic-forgetting law, a population effect, a chat/SFT system, a CORE score, that "deeper is better," that B3 mitigates forgetting, or that P3 confirms P2. Gaps below 0.01 BPB would not have been ranked; here both primary gaps exceed the cutoff with the filed signs.

6.3 Explicit non-reuse of prior confirmatory numbers

Do *not* cite P1.1 native-BPE `test_bpb`=1.164768 as a P3 result. Do *not* cite P2 Gate V English or Tagalog test BPB as P3 outcomes. P3 confirmatory numbers are only the sealed Gate U `val_bpb_full` cells, the filed $C_{\mathrm{tl}}/G_{\mathrm{en}}$ contrasts, and the one authorized Gate V B2-only secondary pair under the P3 tokenizer (1.357842 EN / 2.493197 TL).

6.4 Post-P2 honesty

Because P3 was designed after P2 unblinding, carry-forward budget and cutoff choices are mechanical relative to the P2 apparatus but are not outcome-independent discoveries. Public narratives must retain that disclosure [5].

6.5 Limitations

One seed; Wikipedia-domain text; BOS-best-fit packing crops long documents; reconstructed Tagalog split (original 2019 files unrecovered); no human evaluation; Gate S required a documented clean restart after missing English data; qualitative generation samples, if ever shown, are nonconfirmatory.

7 Conclusion

P3 is a preregistered, post-P2 reverse-direction nanochat continual-pretraining study. Fresh Tagalog TL0 parents cleared P0-T; B0 was frozen; B1/B2/B3 continued under equal budget; validation was sealed before one B2-only secondary test touch. Gate X released $C_{\mathrm{tl}} = 1.023484$ (**observed**) and $G_{\mathrm{en}} = -1.697955$ (**observed**). The contribution is the locked reverse-direction measurement, not a general bilingual forgetting theorem.

Novelty statement

The paper replaces an informal "continue Tagalog nanochat on English" story with a filed reverse-direction design: fresh P3 weights and tokenizer, P0-T before English children, matched B1/B2/B3 budgets, sealed dual-language BPB, and explicit post-P2 disclosure. Empirically, both primary filed patterns were observed in this one-seed apparatus.

Availability

- AsPredicted #307342: `https://aspredicted.org/wd2pc8.pdf`
- ResearchBox #8834: `https://researchbox.org/8834`
- AsCollected: `https://ascollected.org/F36_C2C`
- GitHub: `https://github.com/pageman/nanochat-filipino`
  P3-only trees: `scripts/p3/`, `docs/p3/`, `docs/papers/p3-reverse/`, `docs/run-cards/p3/`, `results/p3/`, `docs/hub/p3-tl-then-en/`
- Hub: `https://huggingface.co/pageman/nanochat-filipino-p3-tl-then-en`
  (B0+B1+B2+B3 together; not on P1.1/P2 Hub repos)

Held-out test text is not redistributed. Secrets and host credentials are not published.

Ethics

Public Wikipedia-derived corpora. No human-subjects experiment. Secrets (host credentials, box passcodes) are not published. Cheng is not a coauthor.

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A Artifact hashes (prefixes OK in body; full here)

Body text uses 8-character prefixes. Full SHA-256 digests:

```
AsPredicted PDF
6cfad0386dff689ad73fa2bf80b70dd4ad191dc44e21e3e4c11c06825ae550b1
Protocol at filing
```

899ba83f0b36f2b4bf4c16b3c675e58788d7763cb439f8a8c3a3c061bda2b986

Tokenizer .pkl
04436b854e0841025a3dd2b46baaeeea07a7ccc252e9f99a19171306f00bc5a8

B0 / B1 / B2 / B3 model_000294.pt
ae621be2c90a3d295f8d21b0e53cb9d4b717803f5d5337fa68f3c3f84d57193c
3f98784bf6e6bdf78785f370140a0db2dd170a848f93897d75c88b44740e2c54
5ee34b20f6601b1753ee6338b5447e091964ddd1087ca66c57434011e6341cc1
521bea166f13a8eee57fef1ac381aa4f715037a3f3f60a85c74c05e02b55ae2d

B3 mix-order
b6ae432b625b6768f84db3f45c411378d1d5a5fdbd15cbfc0e5f6c511196b1a0

TL train / val / test manifests
2b0474c5700dc1eba14def572aa23cc227e4c59c10c2de3ce6b7bda75d137687
4d51644b84d05050bfc8c515079e60f6e437082b6cce2122e9ed00e7b1db2b1c
3bd193458f4c494d84dae345548c0c01cb6cd7275e98d6ed39a41d517a093baf

EN val / test manifests
874dec29844b3d46fc39e5479ee2dc4b3ba37309d9baf3bba4b5654697f3ae3b
2bccabc020cbb8d09273cccdc42ed926957b83824ca767c96fb588041b8d434e

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